

Final Countdown

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This .Rmd provides a skeleton to organize your final portfolio submission. It does **not** include all instructions. Please refer to the assignment details on Canvas to ensure you include everything. You can delete all guiding/instructional text in this .Rmd to avoid crowding, but leave the headers in for organization.

Setting up:

All loading and importing should happen up here at the top of the document. If you need to import a new .csv or .shp, it should be in this section.

```
# load all necessary libraries
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.1.4      v readr      2.1.5
```

```
## v forcats    1.0.0      v stringr   1.5.2
```

```
## v ggplot2    4.0.0      v tibble    3.3.0
```

```
## v lubridate  1.9.4      v tidyr     1.3.1
```

```
## v purrr      1.1.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggthemes)
```

```
library(ggplot2)
```

```
library(sf)
```

```
## Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```
library(maps)
```

```
##
```

```
## Attaching package: 'maps'
```

```
##
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
##      map
```

```

library(dplyr)
library(ggimage)

# import and name your dataset and any other necessary files (e.g. map, images)
mn = st_read("mn_campus")

## Reading layer 'EGISADMIN_BUILDING_POLYGON' from data source
##   '/Users/namgyal/Desktop/R folders/final portfolio/mn_campus'
##   using driver 'ESRI Shapefile'
## Simple feature collection with 393 features and 49 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: -93.25546 ymin: 44.95768 xmax: -93.16829 ymax: 44.9986
## Geodetic CRS:   WGS 84

umn = read_csv("new_umn - Sheet1.csv")

## Rows: 10 Columns: 3
## -- Column specification -----
## Delimiter: ","
## chr (1): location
## dbl (2): lat, long
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

umn_buildings = read_csv("umn.csv")

## Rows: 393 Columns: 33
## -- Column specification -----
## Delimiter: ","
## chr (13): bldg_num, name, legal_name, abbr, campus, address, type, TRI_BLDG...
## dbl (19): TRI_YEAR_0, TRI_YEAR_L, mortenson_dates, gross_sqft, assignable_s...
## date (1): TRI_DATE_0
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

```

Portfolios #1-3

Each portfolio should have a section for the first 3 parts of the portfolio assignment (calibrate, code, and create). The first section for Portfolio #1 below has an outline for what to include in each sub-section.

Portfolio #1

Part 1: Calibrate The research question(s) your plot(s) will answer.

- What are the top 3 buildings with the most men restroom?
- What are the top 3 buildings with the most restrooms by gender?

Part 2: Code For each portfolio, part 2 should include a code chunk for the data wrangling required in that portfolio. To get you started, the first code chunk below should only include any wrangling you need to include in order to create your bar charts from Portfolio #1. Remember to comment your code. Note: if you reuse any datasets you save in previous part 2's, you do not need to repeat the wrangling code in each portfolio's section.

```
top_men = umn_buildings %>% #top_men means top 3 mens restroom.  
  arrange(-(men_restrooms))%>%  
  head(3)  
plot_data = top_men%>%  
  pivot_longer(cols = c(men_restrooms, women_restrooms, undesignated_restrooms), names_to = "gender", va
```

Part 3: Create This section should include all plot code needed to output the graphs you were asked to create for each previous portfolio. These should all be updated and revised to the best of your ability, this is your final portfolio iteration. Remember to comment your code, make sure your plot code is in the order specified by the visualization schema, and update everything you can for clarity.

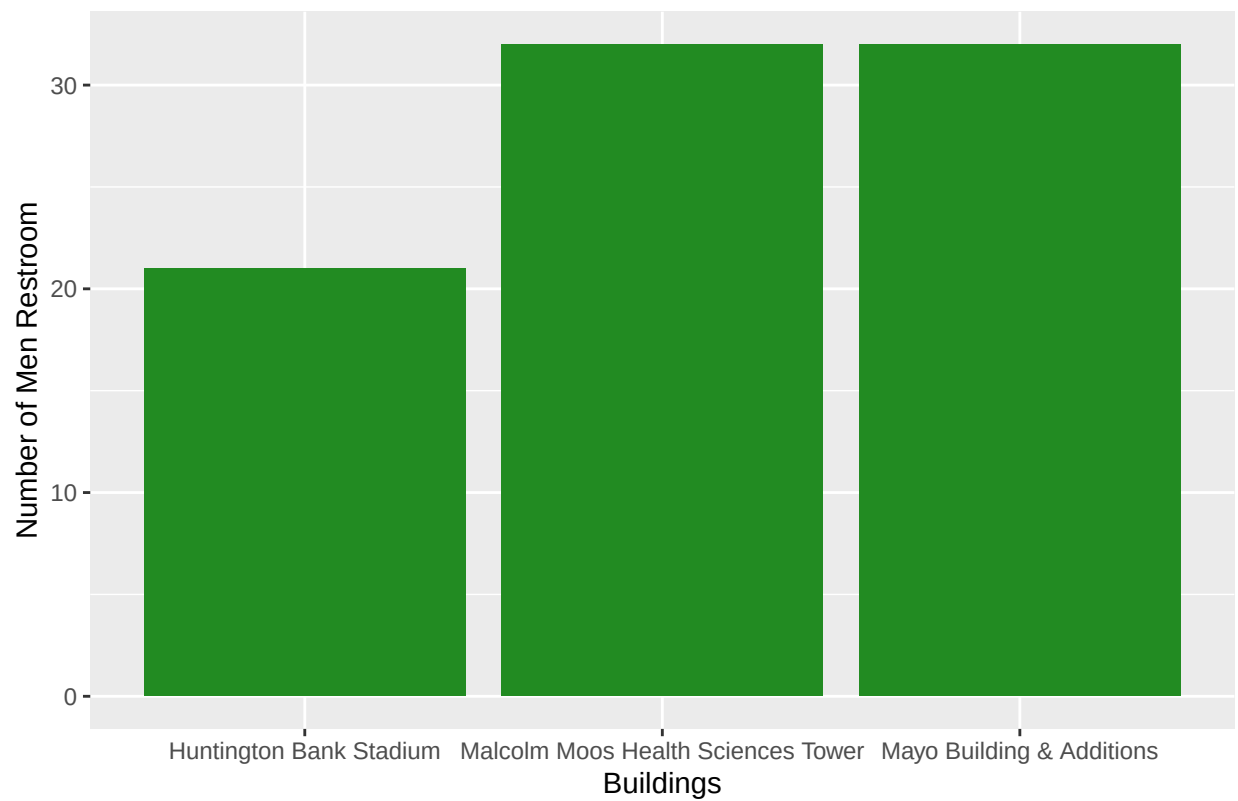
For each portfolio include:

- the plot code and resulting visualization, and
- plot design and schema justification:
 - Why is your code ordered the way it is?
 - Why did you choose to include certain aesthetic mappings and not others?
 - How do your captions and annotations help guide the reader through the story?

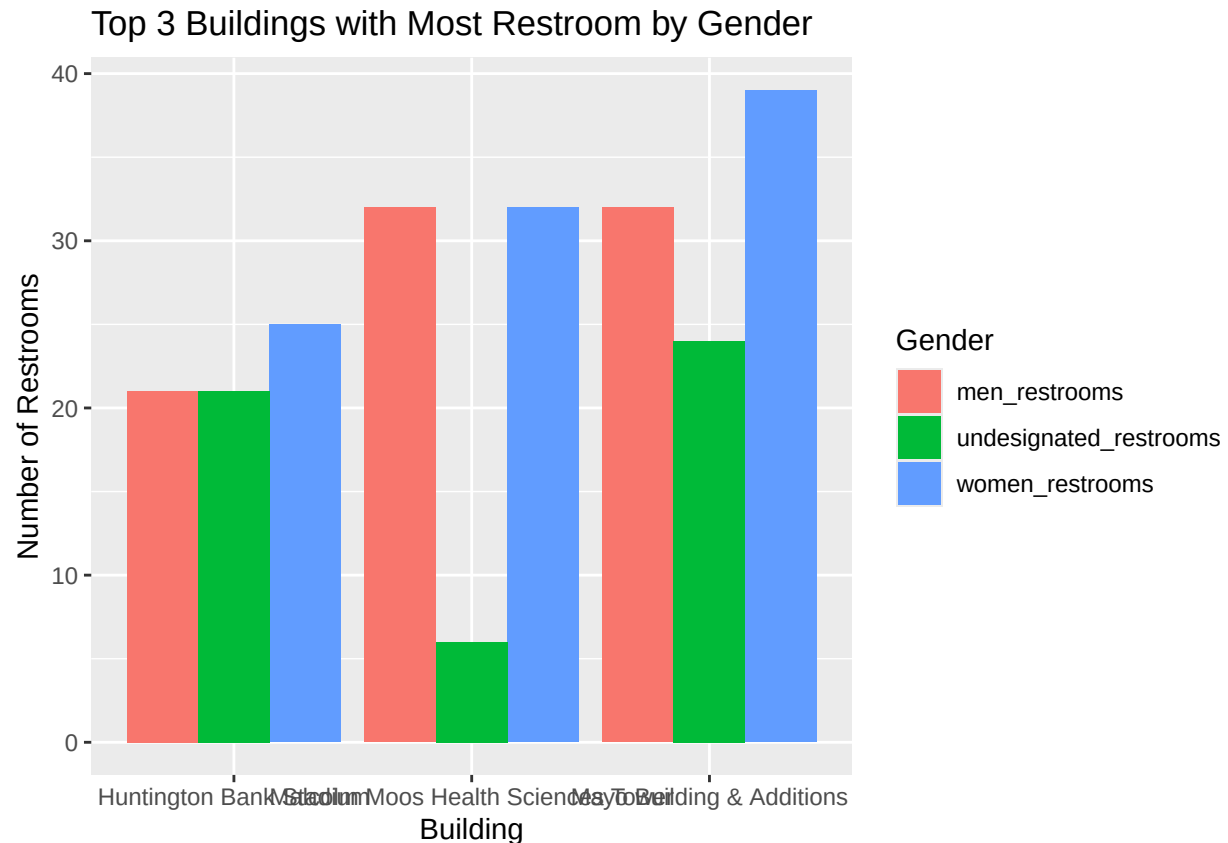
For Portfolio #1, there should be 2 bar charts in this section.

```
ggplot(data = top_men, aes(x = legal_name, y = men_restrooms))+ #top_men is the top three buildings with  
  geom_bar(stat = "identity", fill = "forestgreen")+ #Creates bars whose heights represent actual values  
  labs(title = "Top 3 Buildings With Most Mens Restrooms", #the title.  
        x = "Buildings", #label of x-axis  
        y = "Number of Men Restroom") #label of y-axis
```

Top 3 Buildings With Most Mens Restrooms



```
ggplot(data = plot_data, aes(x = legal_name, y = bathrooms, fill = gender))+ #its the dataset
  geom_bar(stat = "identity", position = "dodge")+ #position = "dodge" puts b
  labs(title = "Top 3 Buildings with Most Restroom by Gender",
        x = "Building",
        y = "Number of Restrooms",
        fill = "Gender") #fill = "gender" is the normal
```



Portfolio #2

Part 1: Calibrate

- What is the distribution of the proportion of accessible buildings across different building types?

```
ada_props = umn_buildings %>%
  group_by(type, ada) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(type) %>%
  mutate(proportion = n / sum(n))
accessible_props = ada_props %>%
  filter(ada == "Accessible")
```

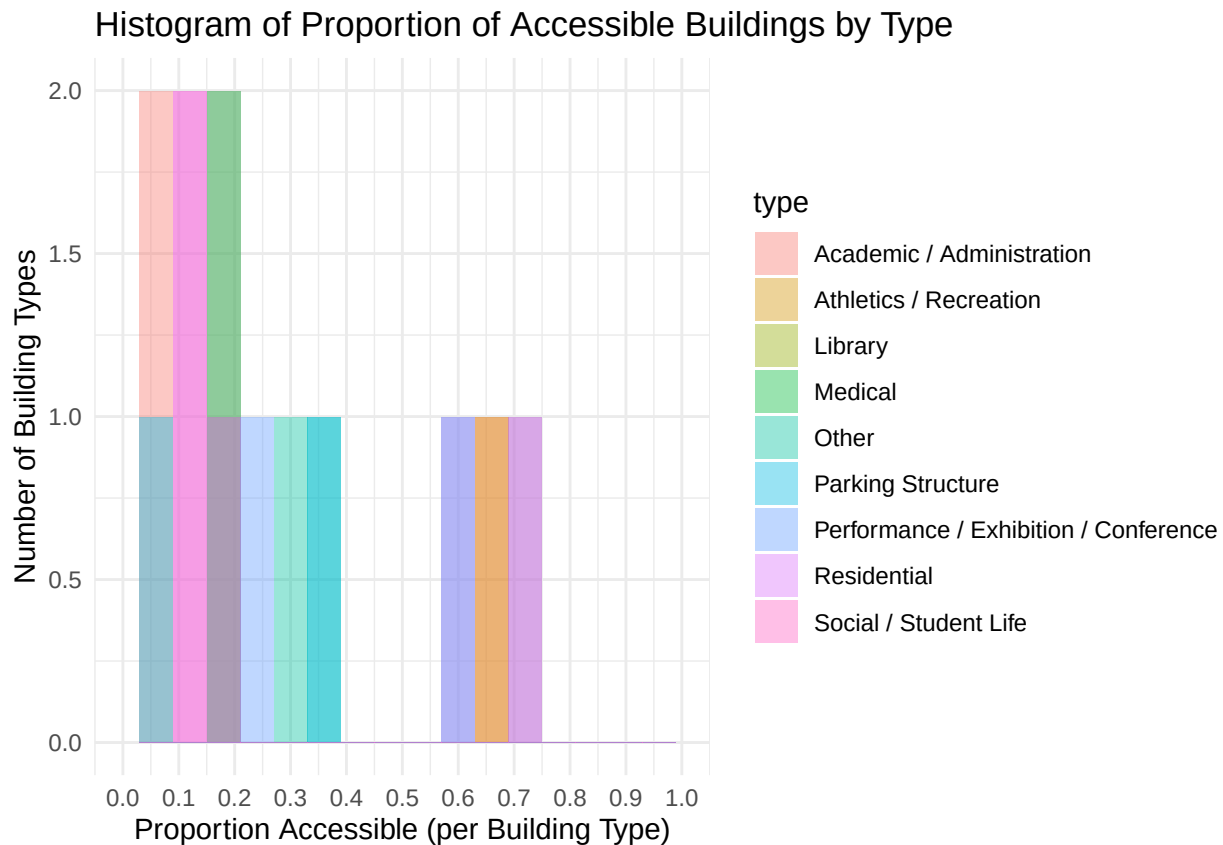
Part 2: Code

Part 3: Create There should be 1 histogram in this section

```
ggplot(ada_props, aes(x = proportion, fill = type)) + # x-axis = proportion of accessible buildings.
geom_histogram(binwidth = 0.06, position = "identity", alpha = 0.40) + #geom_histogram() draws the hist
scale_x_continuous(limits = c(0,1), breaks = seq(0, 1, 0.1)) + #makes x-axis run from 0 to 1
```

```
labs(
  title = "Histogram of Proportion of Accessible Buildings by Type",
  x = "Proportion Accessible (per Building Type)",
  y = "Number of Building Types"
) +
theme_minimal() #makes it look clean with little bit of gridlines
```

```
## Warning: Removed 18 rows containing missing values or values outside the scale range
## ('geom_bar()').
```



Portfolio #3

Part 1: Calibrate

- What is the relationship between the total number of restrooms in a building and the number of men's restrooms?
- Does the relationship between total restrooms and men's restrooms differ by building type?

```
restroom_data = umn_buildings %>%
  mutate(Total_Restrooms = men_restrooms + women_restrooms)
```

Part 2: Code

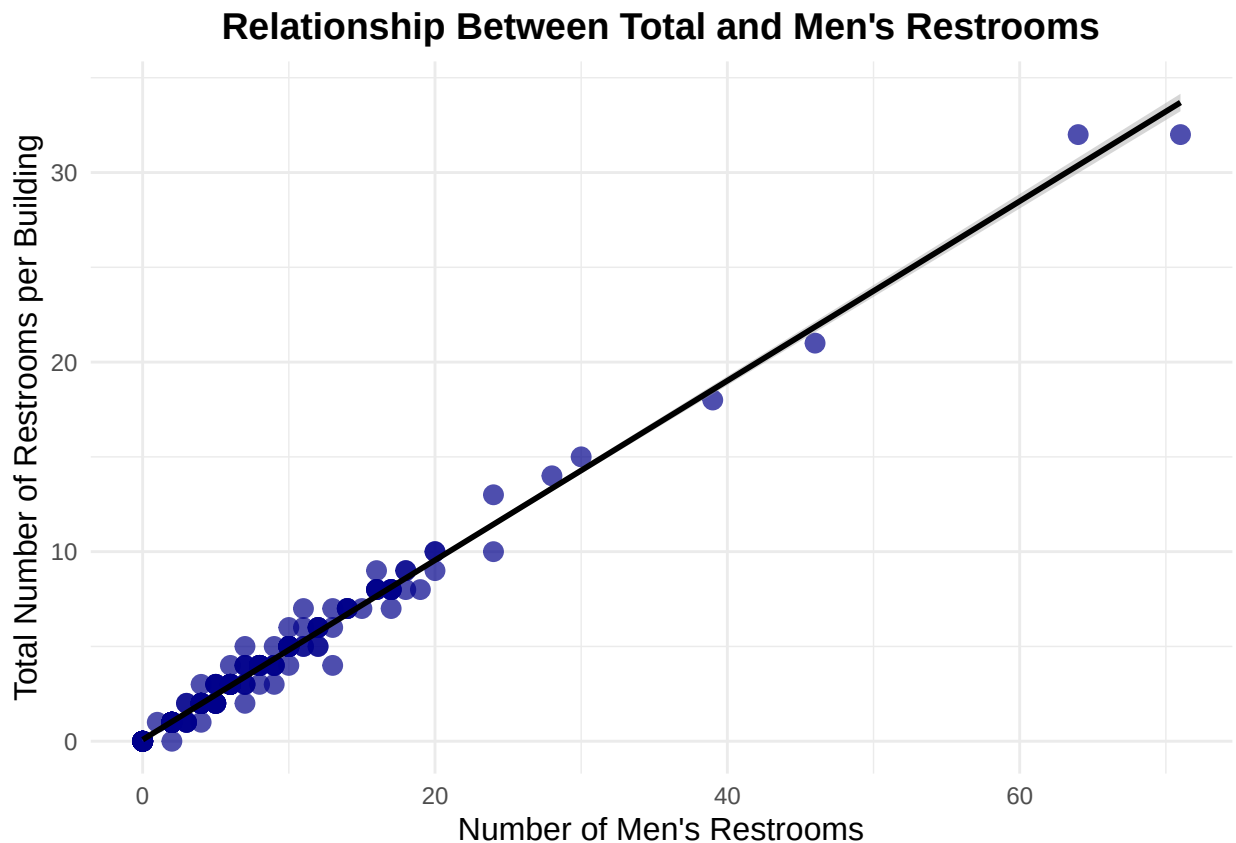
Part 3: Create There should be 2 scatterplots in this section

```
ggplot(restroom_data, aes(x = Total_Restrooms, y = men_restrooms))+  
  geom_point(color = "darkblue", size = 3, alpha = 0.7)+ # makes scatterplot dots.  
  geom_smooth(method = "lm", se = TRUE, color = "black") + #adds a line  
  labs(  
    title = "Relationship Between Total and Men's Restrooms",  
    x = "Number of Men's Restrooms",  
    y = "Total Number of Restrooms per Building") +  
  theme_minimal() +  
  theme(  
    plot.title = element_text(face = "bold", size = 14, hjust = 0.5), ## theme(): Improves reading by c  
    axis.title = element_text(size = 12))
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 169 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

```
## Warning: Removed 169 rows containing missing values or values outside the scale range  
## ('geom_point()').
```



```
ggplot(restroom_data, aes(x = Total_Restrooms, y = men_restrooms, color = type)) + #color = type makes  
  geom_point(size = 3, alpha = 0.7) +
```

```
geom_smooth(method = "lm", se = FALSE) +
facet_wrap(~type) + # Helps compare trends across categories visually
labs(
  title = "Men's vs Total Restrooms by Building Type",
  x = "Total Number of Restrooms",
  y = "Number of Men's Restrooms",
  color = "Building Type") +
theme_light() +
theme(
  plot.title = element_text(face = "bold", size = 14, hjust = 0.5),
  axis.title = element_text(size = 12))
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

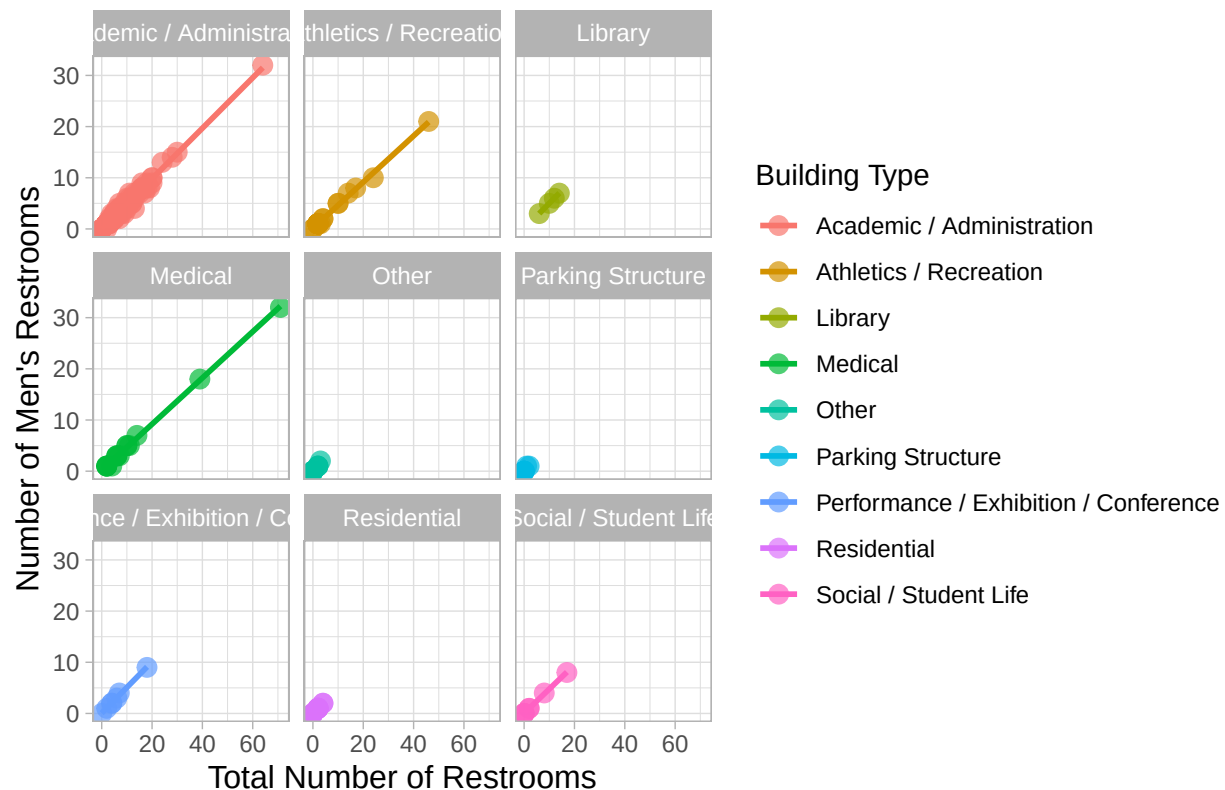
```
## Warning: Removed 169 rows containing non-finite outside the scale range
```

```
## ('stat_smooth()').
```

```
## Removed 169 rows containing missing values or values outside the scale range
```

```
## ('geom_point()').
```

Men's vs Total Restrooms by Building Type



Final Portfolio

Reminder that all instructions and prompting questions are in the assignment details document on Canvas.

Part 1: Calibrate

Reflect on feedback:

- What did the feedback reveal that you hadn't noticed about your visualization or interpretation?
- All the feedback I got, I realized that I didn't fully understand what I was doing, but now that i know what I'm kinda doing it seems pretty easy.

Part 2: Code

Data Wrangling:

```
mn = st_read("mn_campus")
```

```
## Reading layer 'EGISADMIN_BUILDING_POLYGON' from data source
##   '/Users/namgyal/Desktop/R folders/final portfolio/mn_campus'
##   using driver 'ESRI Shapefile'
## Simple feature collection with 393 features and 49 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: -93.25546 ymin: 44.95768 xmax: -93.16829 ymax: 44.9986
## Geodetic CRS:   WGS 84
```

```
umn = read_csv("Sheet1.csv")
```

```
## Rows: 5 Columns: 3
## -- Column specification -----
## Delimiter: ","
## chr (1): location
## dbl (2): lat, long
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

Part 3: Create - Map

Remember to comment your code, make sure your plot code is in the order specified by the visualization schema, and design for clarity.

For each portfolio include the following:

- the research question(s) your plot(s) will answer,
- the plot code and resulting visualization, and
- design and schema justification:
 - Why is your code ordered the way it is? My code is ordered this way because if I don't order it this way the code is not gonna work.
 - Why did you choose to include certain aesthetic mappings and not others? I chose to make it like this because it was the best of my abilities.

- How do your captions and annotations help guide the reader through the story? it will help the readers understand what my thought process was while making this map.

```
ggplot(mn)+ #this is the map of the campus
  geom_sf()+
  geom_point(data = umn, aes(x = long, y = lat, color = "red"))+ #geom_point(data=umn) adds all my clas
  geom_text(data = umn, aes(x = long, y = lat, label = location))+ #it adds the names of the buildings
  geom_image(data = umn %>% slice(6), aes (x = long, y = lat), image = "emoji/goldy-face.png", # this i
size = 0.1) +
  geom_image(data = umn %>% slice(7), aes (x = long, y = lat), image = "emoji/books.png", #this is for
size = 0.1) +
  geom_image(data = umn %>% slice(8), aes (x = long, y = lat), image = "emoji/bed.png", #this is for my
size = 0.1) +
  geom_image(data = umn %>% slice(9), aes (x = long, y = lat), image = "emoji/drama-masks.png", # this
size = 0.1) +
  geom_image(data = umn %>% slice(2), aes (x = long, y = lat), image = "emoji/gold-star.png", #this is
size = 0.3) +
  geom_image(data = umn %>% slice(10), aes (x = long, y = lat), image = "emoji/frut-veg.png", #this is
size = 0.1) +
  geom_image(data = umn %>% slice(2), aes (x = long, y = lat), image = "emoji/marisela.webp", #this is
size = 0.07) +
  coord_sf(xlim = c(min(umn$long) - 0.002, max(umn$long) + 0.002), # this zooms in on the map so everyt
  ylim = c(min(umn$lat) - 0.002, max(umn$lat) + 0.002),
  expand = FALSE)+
  labs(title = "Tenzin's UMN Map",
  x = "",
  y = "",
  color = "classes",
  caption = "source: Tenzin Tharchin")+
  theme_minimal()
```

Tenzin's UMN Map



source: Tenzin Tharchin

Part 4: Communicate

Answer the questions in the “Describe your map” section of Part 4 in the assignment details document.

Across all of the portfolios I created including my bar charts, scatterplots, histograms, and final campus map I always showed the structure and distribution of restrooms and accessibility features across UMN buildings. In each graph, I referenced specific features like color, point position, line trends, and facet layouts. For example, in my faceted scatterplot of Men’s vs. Total Restrooms by Building Type, I saw that buildings categories in Academic/Administration and Medical show a strong upward trend between total restrooms and men’s restrooms. I know this because the linear downwards lines in these facets have a positive slope, meaning that as buildings grow in size, restroom are increases predictably.

Bring it all together:

Answer the questions from the “Big picture; connect your plots to tell a coherent data story” section of Part 4 in the assignment details document.

- Across all my portfolios, I saw how restroom availability, gender, building type, and accessibility is different across campus buildings, usually the ones I use for my classes.
- The main idea is that building type and size strongly influence restroom availability and accessibility, and the buildings where I take classes usually align with these broader campus patterns.
- Buildings with more total restrooms usually have proportionally more to men. Accessibility proportions depends by building type but cluster higher in academic spaces.

- Bar charts showed clear comparisons to top buildings by restrooms. Scatterplots showed relationships between restroom variables. Faceted charts showed differences across building types. The histogram showed distribution of accessibility compliance. The final map added geographic meaning, helping me see how these variables play out across real campus space.
- One thing that surprised me was that buildings with high restroom counts were not always the ones I expected or usually used.
- Why do some large academic buildings have less restrooms than expected?
- New students can use this and see the restroom availability and student navigation patterns.